PRACTICE SHEET

1. How many 3–digit numbers, each less than 600, can be formed from {1, 2, 3, 4, 7, 9} if repetition of digits is allowed?
(a) 216 (b) 180
(c) 144 (d) 120

2. There are four chairs with two chairs in each row. In how many ways can four persons be seated on the chairs, so that no chair remains unoccupied?
(a) 6 (b) 12
(c) 24 (d) 48

3. In how many ways can the letters of the world CORPORATION be arranged so that vowels always occupy even places?
(a) 120 (b) 2700
(c) 720 (d) 7200

4. In fall permutations of the letters of the 'LAGAN' are arranged as in dictionary, then what is the rank of 'NAAGL'?
(a) 48th Word (b) 49th Word
(c) 50th Word (d) 51st Word

5. If a secretary and a joint secretary are to be selected from a committee of 11 members, then in how many ways can they be selected?
(a) 110 (b) 55
(c) 22 (d) 11

12. How many words, with or without meaning can be formed by using all the letters of the word 'MACHINE' so that the vowels occurs only the odd positions?
(a) 1440 (b) 720
(c) 640 (d) 576

13. From 7 men and 4 women a committee of 6 is to be formed such that the committee contains at least two women. What is the number of ways to do this?
(a) 210 (b) 371
(c) 462 (d) 5544

14. If $P(32,6) = k\,C\,(32, 6)$, then what is the value of k?
(a) 6 (b) 32
(c) 120 (d) 720

15. What is the smallest natural number n such that $n!$ is divisible by 990?
(a) 9 (b) 11
(c) 33 (d) 99

16. What is the value of r, if $P\,(5, r) = P\,(6, r-1)$?
(a) 9 (b) 5
(c) 4 (d) 2

17. What is the number of words formed from the letters of the word 'JOKE' so that the vowels and consonants alternate?
(a) 4 (b) 8
(c) 12 (d) None of these